

Unlearning as Distribution Restoration: A Controlled Counterfactual Study, a Validated Selective Screen, and the Limits of Oracle-Free Certification

Sen Yang
Stern School of Business
New York University

sy2576@stern.nyu.edu

Yuen-Hei Yeung
Courant Institute of Mathematical Sciences
New York University

yy@nyu.edu

Abstract

Machine unlearning is commonly evaluated by matching a retrained oracle on trained probes, but in a controlled nonce-fact testbed we find that this criterion can favor methods that still *retain held-out knowledge*: candidates it rates adequate score held-out forget facts -2.82 nats (cluster-CI $[-3.16, -2.48]$) *below* the never-learned level. We recast unlearning as restoration to a **matched retraining reference** and audit both oracle-free screens and certificate-style criteria across 45 model-seed cells spanning five open architecture families. The retraining reference itself **falsifies an absolute retain/round-trip certificate**: even the injected model, which retains R by construction, fails the fixed retain threshold in 41/45 cells (i.e. passes in only 4/45) and its own round-trip in 31/45, and the reference fully certifies in only 1/45, because the thresholds lie below the task’s operating point and the round trip carries a nonzero reacquisition floor that grows with the reacquisition budget. A base-anchored held-out **screen** remains a strong *selective necessary* test: on a sealed challenge suite it rejects the injected model in 45/45 cells and accepts the reference in 44/45, and it partially detects entity-routing suppression through a distributional seam (35/45). A **damage-relative recalibration**, with thresholds set by the reference’s own operating point and the replication noise of independent retraining draws, certifies a small candidate subset in 15/45 cells (concentrated in the noisiest-replication family); where it does not abstain, its picks lie within retraining noise (0.80 nats) on the held-out axes it optimizes, while the common trained-probe criterion sits 5.17 nats away on axes it never optimized (same cells; a supporting comparison, not a head-to-head benchmark). Finally, adversarial challenge models show that *forward-only certification is not sound*: a fixed-magnitude logit suppression attack (-10 on the forget-answer logits) defeats the full forward battery in 12/45 cells (Clopper–Pearson upper bound 39.6%), motivating a finite-query impossibility boundary and scoping our method as an *empirical selective test for methods-as-produced* rather than a universal certificate. An identifiability theorem delimits which facts admit an oracle-free forget threshold at all, with TOFU as the predicted boundary case.

1 Introduction

Unlearning is evaluated by forget-efficacy and retain-utility, but both can be satisfied by *suppression* that diverges from the true retrained model. The retrained (“retain-only”) model is the gold reference but is usually unavailable. We ask two questions: (i) *what does good unlearning look like relative to a matched retraining reference?* and (ii) *can we tell good from bad unlearning without that reference?* A distinctive

feature of our answer is that we turn the audit on our own instruments: we run the retraining reference itself through every criterion we propose, and report which criteria it can and cannot pass.

Our contributions are:

1. **Protocol critique.** In a controlled dose-matched injection testbed with a matched counterfactual retraining reference, the common trained-probe oracle-KL criterion *rewards residual knowledge*: candidates it rates adequate score held-out forget facts -2.82 nats (cluster-CI $[-3.16, -2.48]$) below the never-learned level, and its preferred selections sit 5.17 nats from the reference’s held-out position.
2. **Absolute certification falsified by its own reference.** Fixed retain/round-trip thresholds sit below the injected model’s operating point (it fails them in $41/45$ and $31/45$ cells); the retraining reference certifies in $1/45$. The round-trip floor is largely reacquisition-budget drift (three of five families close their own loop at short budgets), with one family showing intrinsic path-dependence.
3. **A validated selective screen.** A base-anchored held-out rank screen, audited on a sealed challenge suite of known-label models: it rejects the injected model in $45/45$ cells, accepts the reference in $44/45$, and partially detects entity-routing suppression through a distributional seam ($35/45$). It is a *necessary* test with measured sensitivity, not a sufficiency certificate.
4. **Damage-relative recalibration.** Re-anchoring thresholds to the reference’s own operating point, with tolerances set by the replication noise of independent retraining draws, yields a selective partial positive: certification in $15/45$ cells, abstention elsewhere, and non-abstaining picks within retraining noise of the reference on held-out axes (0.80 vs 5.17 for the trained-probe criterion on the same cells; a supporting comparison, not a head-to-head benchmark, since the selector optimizes the recalibrated axes).
5. **An adversarial boundary.** On the same challenge suite, a fixed-magnitude logit suppression attack defeats the full forward battery in $12/45$ cells (Clopper–Pearson upper bound 39.6%): forward-only certification is not adversarially sound, so our guarantee is scoped to *methods-as-produced*, and a finite-query impossibility argument explains why.
6. **An identifiability theorem** delimiting when oracle-free selection is possible at all, with TOFU as its predicted boundary, plus a finite-sample certifiability bound on any rank-test forget screen.

2 Controlled Testbed with a Matched Retraining Reference

Injection. From a base model M_0 , we continue-pretraining to co-acquire a **forget set** F and a **retain-acquired set** R of nonce facts (dose-matched; wikitext retain mix), yielding M_{inj} . We reuse the controlled nonce-injection *substrate* introduced for acquisition-diversity experiments; the present paper studies a distinct evaluation problem (whether unlearning candidates restore the matched retraining distribution) with a different intervention variable, target estimand, and failure mode. To avoid double-counting across that companion work, we note that the shared element is the injection methodology only: the 45 model-seed cells, the 820 unlearning candidates, the forget/retain facts, and the held-out and sealed challenge (AUDIT; Section 6) probe pools of this paper are constructed for restoration evaluation and are not reused as, nor counted as, replications of any acquisition-diversity result (though both lines use `gate_proj` masks, here as one *unlearning* candidate family, there as a *deletion-cost* instrument).

Matched retraining reference. M_{oracle} replays the identical stream **without** F (inject R only): a model that never saw F but has the same retain competence. We call it a *matched retraining reference* rather than an “exact oracle”: bf16 training is nondeterministic, and rebuilding the reference from independent draws shifts per-fact held-out deltas by up to $\delta_{\text{equiv}} = 0.93$ nats at the P90 (per-family range 0.82 – 2.92 ; Section 5). Any single draw is therefore one realization of retraining, not a unique counterfactual, and δ_{equiv} is the scale below which two retrainings are indistinguishable, a quantity the rest of the paper uses repeatedly.

Why mixed acquisition. If the stream contained only F , full checkpoint-rollback ($= M_0$) would trivially “forget”; co-acquiring R makes rollback destroy retain (retain-NLL 6.6 vs oracle 2.4), so a method must do real work.

Metrics. *trained-probe oracle-KL* = $\text{KL}(\text{candidate} \parallel M_{\text{oracle}})$ on $F+R$ *trained* probes, the criterion the field uses as ground truth, which Section 3.1 shows can reward residual knowledge; *held-out* forget-NLL on F paraphrases never used in any training or unlearning loss (the surface our screens operate on); retain-NLL on R ; general damage on wikitext. We keep these roles separate throughout: the trained-probe criterion is *audited*, not assumed; held-out deltas anchored to M_0 are the primary evaluation surface; and forget-NLL / retain-NLL alone are diagnostics a suppression method can also satisfy.

Models. Qwen2.5-0.5B/1.5B, SmolLM2-360M/1.7B, Pythia-410M/1.4B, GPT-2-medium/large, OLMo-2-1B (five families; SwiGLU and GELU). Real unlearning methods (below) are architecture-agnostic; the `gate_proj` mask variant applies only to SwiGLU models. We deliberately span diverse small-to-mid open families rather than a single large model: the audit requires a *matched retraining reference* (and its independent redraws) and a re-acquire round trip for *every* model-seed cell (45 cells), which is feasible at this scale and gives cross-architecture evidence; scaling the evaluation to larger instruction-tuned models is future work. TOFU (on Phi-1.5) is included not as a comprehensive real-world unlearning sweep but as the predicted *boundary* case for identifiability (Section 9).

3 Distribution Restoration as the Unlearning Target

Methods compared. Each starts from M_{inj} , at matched compute, with full forget/retain Pareto fronts: path-specific **KL reversion** (forget $\rightarrow M_0$, retain $\rightarrow M_{\text{inj}}$), **task-vector negation** ($M_{\text{inj}} - c \cdot (M_{F\text{only}} - M_0)$), `gate_proj` **mask** reversion; versus **gradient ascent (GA)** and **NPO** (Zhang et al., 2024), each +retain; plus **checkpoint rollback** ($= M_0$).

Result (controlled, matched forgetting). Restoration methods (KL-reversion, task-vector) reach oracle-KL $\approx 2\text{--}4$; suppression (GA/NPO) $\approx 5\text{--}7$ even after a **32-configuration fair sweep** ($\text{lr} \times \beta \times \text{retain-weight}$): they over-forget (forget-NLL far past the oracle) and collapse retain (retain-NLL $\rightarrow 0 \ll$ oracle), i.e. they *distort* rather than restore. Checkpoint rollback destroys R . Across 3 seeds, at matched forgetting restoration is $\sim 2\times$ closer to the oracle than suppression (Table 1).

Interpretation. Deletion that approximates the retrained model is *restoration* of the pre-acquisition distribution on F while preserving the post distribution on R , not indiscriminate loss-increase.

3.1 The trained-probe criterion rewards residual knowledge

The diagnostic above uses proximity to the retraining reference as measured on *trained* probes: the phrasings that appeared in injection and unlearning losses. Auditing that criterion against *held-out* paraphrases exposes a systematic failure: across the 45-cell matrix, candidates the trained-probe criterion rates adequate score held-out forget facts a mean of **-2.82** nats (cluster-bootstrap CI by cell $[-3.16, -2.48]$, entirely below zero; per-fact median -2.91 , $n = 948$) *below* the never-learned level. They demonstrably retain paraphrase-recoverable knowledge of F while matching the reference on the phrasings the criterion happens to query. Matching the retrained model on trained probes is therefore *not* evidence of restoration. That observation motivates every held-out, base-anchored instrument in the rest of this paper.

The method dichotomy. The same held-out surface splits the method space cleanly. Loss-based methods (KL-reversion, GA, NPO) suppress only the phrasings in their loss: 77–87% of their candidates are excluded by the held-out screen as *retaining* F . Weight-space methods (`gate_proj` mask, task-vector) generalize forgetting across phrasings but damage retain. Under the absolute (fixed-threshold) bar (screen $\wedge$ retain $\leq 0.5 \wedge$ round-trip ≤ 0.5), only 3/820 candidates matrix-wide satisfy all three, an outcome that Section 5 traces to the bar itself rather than to the methods.

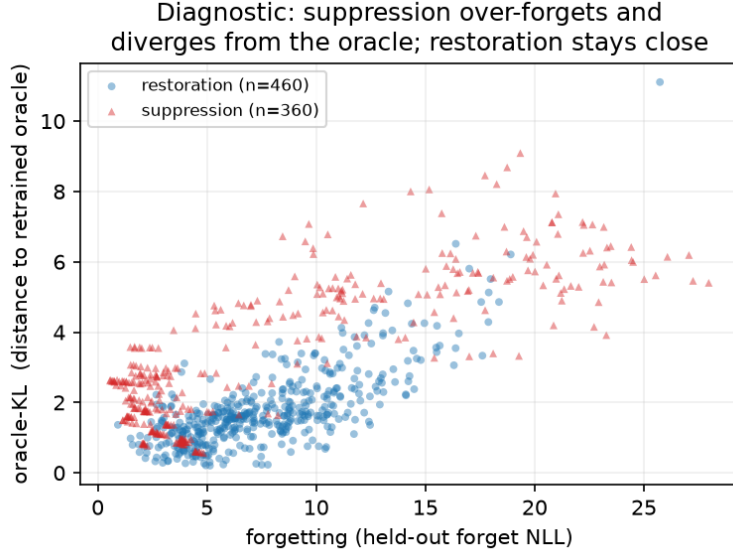


Figure 1: Diagnostic (pooled candidates, 45 matrix cells). Suppression methods (GA/NPO) over-forget *and* diverge from the matched retraining reference (high trained-probe KL); restoration methods (task-vector, KL-reversion) stay close to the oracle across the forgetting range.

Table 1: Diagnostic: oracle-KL of restoration vs suppression methods at matched forgetting, in the controlled testbed (across 3 seeds and a 32-configuration fair sweep) and on TOFU’s real retrained reference. Lower is closer to the retraining reference. Restoration stays $\sim 2\times$ closer to the oracle throughout.

setting	restoration (KL-reversion, task-vector)	suppression (GA, NPO)
controlled (oracle-KL)	$\approx 2-4$	$\approx 5-7$
TOFU (oracle-KL)	≈ 0.1	0.7–1.2

4 Oracle-Free Round-Trip Certificate

Idea. A genuine counterfactual deletion closes the loop *acquire* $\rightarrow$ *delete* $\rightarrow$ *re-acquire*: after re-acquiring F , a truly-restored model returns to $\approx M_{\text{inj}}$ on retain; a suppression model’s collateral damage **persists**. We define the oracle-free **round-trip residual**

$$\text{cert}(C) = \text{KL}(\text{reacquire}_F(C) \parallel M_{\text{inj}}) \text{ on retain} + \text{wikitext},$$

which uses only M_{inj} and the forget data, with no oracle.

Forget-qualification: the base-anchored screen (necessary). The round-trip residual ranks *restoration quality*; it must be paired with a check that the candidate actually forgot. The principled form is an **under-forgetting exclusion test** on *base-anchored deltas*, and we state explicitly which model scores what, because two subtly different designs are wrong. For candidate C , forget fact i , and never-learned probe fact j (facts of the same nonce type from unused seeds, never injected anywhere), define

$$\Delta_i^F = \text{NLL}_C(F_i) - \text{NLL}_{M_0}(F_i), \quad \Delta_j^P = \text{NLL}_C(P_j) - \text{NLL}_{M_0}(P_j),$$

both on *held-out* paraphrases and both scored by the *candidate itself*, anchored to the base model M_0 . A one-sided exact rank (Mann–Whitney) test of $\{\Delta_i^F\}$ against $\{\Delta_j^P\}$, Holm-corrected at $\alpha = 0.05$ over the candidate pool of the cell, excludes candidates whose forget deltas are significantly *below* their own probe deltas (residual knowledge). Anchoring to M_0 matters twice over: M_0 never learned F or P , so it is symmetric

in the two sets and oracle-free, and the per-fact anchoring blocks two failure modes of naive designs. Scoring probes with M_{inj} compares NLLs *across models*, so global damage masquerades as forgetting, while anchoring deltas to M_{inj} injects the very F -vs- P label under test (its baseline gap between learned F and never-seen P is the label). Probes and the retraining reference are never used to score candidates.

When the check is possible at all. This test is usable only if the design supplies enough probes, and the constraint is combinatorial rather than statistical:

Proposition 1 (Finite-sample certifiability). *Fix a cell with K candidates, m forget items, and n matched never-learned probes. For the one-sided exact rank test, the smallest attainable p -value is $1/\binom{m+n}{m}$. Holm’s step-down rejects its most significant hypothesis only if that p -value is at most α/K . Hence no candidate can be excluded, at any effect size, unless*

$$\binom{m+n}{m} \geq K/\alpha.$$

The bound is a property of the test’s null lattice, not of the data: it holds however badly a candidate under-forgets. With $\alpha = 0.05$, our pools require $\binom{m+n}{m} \geq 400$ in SwiGLU cells ($K = 20$) and ≥ 320 in GELU cells ($K = 16$, which lack the `gate_proj` mask arm). At $m = n = 4$, $\binom{8}{4} = 70$ falls short of both, so the exact untied screen is **vacuous by construction** (it excludes 0 of 820 candidates). The matrix therefore uses $n = 16$ probes at the same dose ($m = 4$): $\binom{20}{4} = 4845$ clears both pools, the attainable floor is $1/4845 \approx 2.06 \times 10^{-4} < \alpha/K$, and the powered screen excludes 498 of 820 candidates. We emphasize the correct reading of the bound: it guarantees attainable p -value *resolution* sufficient for Holm rejection, not power against a specified effect size. TOFU supplies $m = n = 40$, where $\binom{80}{40} \approx 10^{23}$ clears the bound by twenty orders of magnitude, and the screen is decisive there (16/16 candidates excluded in every seed).

Probes are the free knob. Read as a design rule, the bound separates two costs that are easy to conflate. Enlarging the forget set raises m , but it also changes the injection dose, so it perturbs the very system under study. Enlarging the probe set raises n without touching that system: never-learned probes enter only as forward passes against M_{inj} , and never enter injection, unlearning, or the residual $\hat{c}$. Raising n is not statistically inert: a larger probe set changes the sampled reference distribution the forget losses are ranked against, so it changes the comparison and not merely the resolution of the lattice. Even so, it leaves the intervention, the candidates, and every candidate-side quantity untouched. If probes are matched one-per-forget-item ($n = m$), the requirement $\binom{2m}{m} \geq K/\alpha$ forces $m \geq 6$; holding our dose fixed at $m = 4$, $n = 8$ is the smallest probe count sufficient in both pools ($\binom{11}{4} = 330$ clears 320 but not 400; $\binom{12}{4} = 495$ clears both); we adopt $n = 16$, comfortably above the minimum, and store per-item values so every $n \leq 16$ sub-panel is recoverable by subsetting. The asymmetry is the practical content of the proposition: a screen can be underpowered purely because probes were treated as scarce when they were free.

Implementation audit. Two details make the bound auditable rather than nominal. First, the exact lattice is what our implementation actually evaluates in the regime the bound is about: we verified exactness at $m = 4$ for every $n \leq 16$ we use, and in the matched case up to $m = 8$, beyond which it substitutes a normal approximation. At $m = 4$, $n = 16$ the attainable minimum is exactly $1/\binom{20}{4} = 2.06 \times 10^{-4}$, which is also the smallest p -value observed anywhere in the 820 candidates, attained by 238 of them. This is a data-level audit of the lattice: the test operates on *fact-level* scalars (held-out template NLLs are averaged *within* each fact before ranking), and a paired per-fact sign design, whose floor is $2^{-4} = 0.0625$, could not produce these p -values. Second, ties are the one caveat: they push the implementation onto the approximation, which is *not* floored by $1/\binom{m+n}{m}$, so in tied cases the bound constrains the exact test rather than the code path taken; across all 820 candidates no tied case attained a smaller minimum. TOFU’s $m = n = 40$ likewise falls in the approximation’s regime, so its $p \approx 7 \times 10^{-15}$ is the approximation’s rather than the exact lattice’s. It is far below α/K either way, so the screen’s decisiveness there does not rest on the exact argument.

Selective certification (abstention). We therefore treat certification as *selective*. The selector returns a certified choice only when the forget-qualification threshold is identifiable from the never-learned reference (Section 9), when Proposition 1 is satisfied so the screen can reject at all, and when a candidate clears

the reference-relative bar of Section 7; otherwise it reports **uncertified** rather than silently degrading to residual-only ranking. The failure modes are distinct and we observe each: TOFU fails *identifiability* (the reference is unreachable, so *every* candidate is excluded; Section 8); the underpowered $n = 4$ design fails *power* (resolved at $n = 16$); and the recalibrated bar abstains in 30/45 matrix cells. A certificate that cannot say “uncertified” would report these cases as success.

Screen-pass is not restoration (equivalence). Because the screen is one-sided, passing it means only “not demonstrably retaining”: non-rejection is not equivalence to the never-learned state, and with a weak test it would be vacuously easy. We quantify the gap. Among the 322 screen survivors of the matrix, the median absolute gap between forget and probe deltas is **1.93** nats (cell-clustered bootstrap CI [1.68, 2.34]), well above the retraining-replication tolerance $\delta_{\text{equiv}} = 0.93$; only 28.0% of survivors (CI [22.6%, 34.0%]) are TOST-equivalent to never-learned at that tolerance. The screen is a *necessary* filter, and the recalibrated bar of Section 7 therefore adds an explicit equivalence axis rather than treating survival as success.

Selector (oracle-free, at a glance). Given candidate unlearned models $\{C\}$ (the *delete* outputs of the methods under comparison), each starting from M_{inj} :

1. **Re-acquire:** fine-tune each C on the forget set F to convergence, giving $\text{reacquire}_F(C)$.
2. **Measure residual:** $\text{cert}(C) = \text{KL}(\text{reacquire}_F(C) \parallel M_{\text{inj}})$ on retain + wikitext.
3. **Qualify:** exclude any C that fails the under-forgetting test against matched never-learned probes (residual knowledge).
4. **Select:** among survivors, return $\arg \min_C \text{cert}(C)$.

No oracle, gold posterior, or probe labels enter the selection; only M_{inj} and the forget data are used.

Selector cost. The certificate’s cost is a per-candidate re-acquisition fine-tune (each C is briefly fine-tuned on F to convergence), so selector compute scales linearly in the number of candidates times the re-acquisition cost; it uses no retrained oracle and no held-out probes at selection time. This makes it suited to offline method-selection studies rather than per-query deployment.

Scope of the guarantee (what “certificate” does and does not mean). We use *certificate* in a deliberately narrow, selection-theoretic sense, and state the scope explicitly because the term is easily over-read. (i) *It is a selector, not a deletion or privacy guarantee:* it ranks a *fixed, given pool* of candidate unlearned models and says which one is closest to the counterfactual retrained model; it certifies nothing about an individual model in isolation, and makes no differential-privacy or right-to-erasure claim. (ii) *Offline setting:* the evaluator holds M_{inj} , the forget data F , and the candidate pool, and may fine-tune; the retrained oracle is never available. (iii) *Non-adaptive methods:* candidates are produced without knowledge of the certificate. A bilevel adversary that optimizes a candidate *against* the round-trip residual is outside the present analysis and is the most natural attack on the construction (Section 11). (iv) *Assumption-bound:* oracle-consistency holds under the threshold-separation and residual-soundness assumptions stated in Section 9, and only where the forget-qualification threshold is identifiable from the observable never-learned reference; the finite-sample test is an under-forgetting screen, not a proof of adequacy for any single candidate.

5 The Matched-Retraining Audit: the Absolute Bar Fails Its Own Reference

The natural certificate pairs the screen with absolute thresholds (retain damage ≤ 0.5 nats and round-trip residual ≤ 0.5), and under that frozen bar only 3/820 candidates certify matrix-wide (Section 3.1). The bar must first survive its own reference: we run **the retraining reference, and the injected model, through the full battery, component-wise, in every cell.**

Component-wise audit of the reference and injected model. The reference passes the base-anchored screen in 43/45 cells, the one component it reliably clears. It passes *retain* ≤ 0.5 in only 11/45 and the round-trip threshold in 7/45; it fully certifies in **1/45** cells. The diagnosis is not that retraining is poor but that **the**

thresholds sit below the task’s operating point: M_{inj} itself, which retains R perfectly by construction, passes the absolute retain threshold in only **4/45** cells (median retain-NLL 1.20, range 0.24–4.45: models simply do not memorize the nonce retain facts below ~ 1 nat at this dose), and M_{inj} ’s *own* round-trip self-closure (re-acquiring F into the model that already knows $F+R$ and measuring the KL back to itself) exceeds 0.5 in **31/45** cells (median floor 0.92). A candidate was being asked to pass thresholds that the model it restores *toward* cannot pass. Under a consequence table frozen before the run, this outcome retires the absolute bar: “no method certifies” was a statement about the bar, not the methods.

Replication noise of the reference. Rebuilding the reference from three independent draws in one representative cell per family gives the per-family indistinguishability tolerance δ_f (P90 per-fact held-out delta gap): 0.82–2.92 nats across families, median 0.93. Draw-to-draw spreads of retain-NLL and round-trip residual supply the reference-noise margins ε_f used below. The wide family spread is itself a finding: a single global tolerance mis-scales at least some families, so all reference-relative results below are computed per family, with a global-minimum sensitivity row.

The round-trip floor is largely budget drift. Re-measuring M_{inj} ’s self-closure at reacquisition budgets $\{15, 30, 45, 75\}$ steps shows the floor rising near-monotonically with budget in every family; at the shortest budget three of five families close their own loop below 0.5 (gpt2 0.05, SmolLM2 0.12, Pythia 0.44), while OLMo-2 never does (floor 1.4–2.1 at every budget: intrinsic path-dependence). The round-trip residual at a frozen budget is therefore dominated by *reacquisition drift* common to every model put through the protocol. This is why the recalibration below subtracts the per-cell floor: candidates and floor share the budget, so the subtraction removes the common-mode drift.

6 A Validated Selective Screen and Its Adversarial Boundary

A screen worth trusting must be validated as an instrument: measured true-accept and false-accept rates on models whose ground truth is known by construction. We freeze a sealed AUDIT template pool (never used in any loss, calibration, or selection; used exactly once, here) and pass a known-label challenge panel through the battery blind, in every cell: the base model M_0 ; the injected model M_{inj} ; the retraining reference; **logit suppression** (an additive -10 penalty on the forget-answer token logits of M_{inj} : pure suppression with the knowledge intact); an **entity router** (dispatch inputs containing a forget-subject token sequence to M_0 , all else to M_{inj} ; suppression that generalizes across phrasings); **embedding corruption** (zeroing the forget subject/object embedding rows); and weight-space interpolations $\lambda M_{\text{oracle}} + (1-\lambda)M_{\text{inj}}$.

Sensitivity and true-accept. The screen rejects M_{inj} in **45/45** cells and accepts the reference in **44/45**; acceptance along the interpolation path rises monotonically toward the reference. On the corrected screen the residual-based selection also never picks a suppression method in the matrix (0/45; exact one-sided binomial upper bound 6.4% per-cell, 28.3% clustered by the nine architectures; we report the bound, not “never”).

A distributional seam partially detects routing. The entity router is behaviorally reference-like on every queried input, yet the screen rejects it in **35/45** cells. The mechanism is a seam the router cannot avoid: its forget-side responses come from M_0 (deltas exactly zero) while its probe-side responses come from M_{inj} , whose drift on never-learned probes is nonzero, and the base-anchored rank test reads that inconsistency as residual structure. We call this *base-delta drift-seam sensitivity*: the screen is strictly stronger against routing than a naive paraphrase test, though the router still passes in 10/45 cells, a concrete, quantified witness for the finite-query impossibility argument of Section 9.

The adversarial boundary: forward-only certification is not sound. Logit suppression exposes the limit. The screen *alone* false-accepts it in 28/45 cells (by design: a one-sided under-forgetting test cannot penalize looking *more* forgotten), and the equivalence axis of Section 7 catches most of those; but in **12/45** cells the fixed-magnitude penalty happens to land the forget-answer NLL within the family tolerance of the reference’s position, and the *entire forward battery* accepts a model whose knowledge is intact. We report 12/45 descriptively and by family (Qwen and OLMo cells dominate the evasions): the 45 cells are fixed,

Table 2: Selection quality on identical held-out axes, averaged over the 15 cells where the recalibrated certificate does not abstain (same-cell comparison). feq: distance of the pick’s held-out forget position from the same-draw reference (nats); damage axes clipped at 0; trained-probe oracle-KL shown for reference (the criterion Section 3.1 audits). The recalibrated pick lies within retraining noise ($\text{feq} < \delta_f$) where the trained-probe criterion’s preferred candidate retains residual held-out knowledge 5.17 nats from the reference’s position.

selector (same 15 cells)	feq	retain-dmg ⁺	cert-dmg ⁺	trained-probe KL
recalibrated certificate pick	0.795	0.729	0.767	3.555
absolute-bar pick	3.298	0.990	0.166	2.103
trained-probe best	5.167	0.838	0.312	1.551

family-clustered, and heterogeneous, so this is not an i.i.d. success rate; the one-sided Clopper–Pearson upper bound (39.6%) characterizes *these* cells, not a guarantee about future models. Catching such an attack requires the relearning axes (the suppressed model relearns instantly) or white-box access, neither available to a forward-only protocol. Embedding corruption, by contrast, is rejected almost everywhere on retain damage (compatible with M_{inj} in only 3/45 cells): in a nonce-token testbed, deleting embedding rows is a blunt distributional intervention, not surgical suppression. The consequence is stated plainly in every claim of this paper: our certification is an **empirical selective test for methods-as-produced**, not an adversarially sound certificate.

7 Damage-Relative Recalibration: a Selective Partial Positive

The audit’s constructive consequence is a recalibrated bar whose rules were frozen before the analysis ran: measure candidates *relative to the reference’s operating point*, at the tolerance retraining itself cannot beat. A candidate C in a cell of family f is CERTIFIED iff (i) it passes the base-anchored screen; (ii) *forget equivalence*: $|\text{med}(\Delta^F(C)) - \text{med}(\Delta^F(\text{reference}))| \leq \delta_f$, against the *same-draw* reference arrays of that cell; (iii) *retain damage* $\text{NLL}_R(C) - (\text{NLL}_R(M_{\text{inj}}) - \varepsilon_f) \leq \delta_f$; and (iv) *round-trip damage* $\text{cert}(C) - (\text{floor} - \varepsilon_f) \leq \delta_f$, where cross-draw references are shifted against the candidate by the reference-noise margins ε_f . A degeneracy gate excludes candidates “retaining better than M_{inj} ” by more than noise (collapse onto R , not restoration). The relearn-veto axis of the original bar is unavailable retrospectively (stored candidates carry no relearning trajectories), so this is a *veto-less retrospective* certificate, stated as such.

Result. 38/820 candidates certify, in 15/45 cells; the certificate abstains in the remaining 30. The abstention is selectivity by design, not failure. The `gate_proj` mask family leads (16/100 candidates, 16%), with task-vector, NPO, and GA at 4% and full-parameter KL-reversion at 0% (it retains held-out F ; the screen removes it). Certification volume is tolerance-sensitive and we disclose its concentration: the Qwen family, whose retraining replication is noisiest ($\delta_f = 2.92$), contributes 29 of the 38; under a global conservative tolerance ($\delta = \min_f \delta_f = 0.82$) the count falls to 4 candidates in 4 cells. The robust claim is therefore not the count but the selection quality below.

Selection quality (same-cell). Table 2 compares three selectors on the same cells and the same held-out axes. The comparison against the trained-probe criterion is non-circular (that selector never optimized these axes), and it lands 5.17 nats from the reference’s held-out position, versus 0.80 for the recalibrated pick: *within* the family’s retraining noise. The two criteria disagree sharply about which candidates are good (the recalibrated picks score *worst* on trained-probe KL), and Section 3.1 says which to trust. We are deliberately careful about what this table does and does not show. That the recalibrated pick scores best on the recalibrated axes is *partly definitional* (the selector optimizes those axes), so we do **not** present 0.80 vs 5.17 as a head-to-head selection benchmark, and a standalone selection claim would require a second, untouched held-out probe suite with one candidate frozen per cell *before* that suite is scored (future work). The two claims that survive this caveat, and that we do make, are narrower: (i) the common trained-probe criterion selects candidates that sit 5.17 nats from the reference on held-out axes *it never optimized* (a

property of that criterion, not a contest), and (ii) in its non-abstaining cells the recalibrated selector’s pick sits, in absolute terms, within the family’s retraining-replication tolerance δ_f . The recalibration is thus a *supporting* selective positive under the audit-led framing, not the paper’s load-bearing result; the falsification and screen-validation contributions stand without it.

8 External Validation on TOFU

Setup. TOFU (Maini et al., 2024) forget10/retain90 on Phi-1.5; $M_0 = \text{phi-1.5}$, $M_{\text{inj}} = \text{tofu_ft}$ (all TOFU), $M_{\text{oracle}} = \text{the official tofu_ft_retain90 retrained reference (never saw forget10)}$.

Diagnostic holds on the real oracle. Restoration (KL-reversion) oracle-KL ≈ 0.1 vs suppression 0.7–1.2; at matched forgetting restoration is both closer to the oracle **and** assigned a lower round-trip residual (6/7 pairs). The core scientific claim replicates outside the synthetic testbed.

Certificate boundary: the selector abstains, and should. TOFU is where oracle-free *qualification* fails, and the failure is instructive rather than incidental. TOFU facts are *style-inferable*: the retrained oracle itself only reaches forget-NLL ≈ 0.5 –2, while matched never-learned probes (novel QA) sit at NLL ≈ 8 . No achievable forgetting reaches the never-learned reference, so the exclusion test rejects *every* candidate, in all three seeds (16/16 candidates excluded per seed; smallest $p \approx 7 \times 10^{-15}$).

This is the opposite failure mode from the controlled matrix, and the two together delimit when certification is possible at all. Here the test has ample power ($m=n=40$, so Proposition 1 is satisfied by a wide margin) and still excludes everything, because the *reference itself* is unreachable: identifiability fails. In the matrix the reference is informative but the test has no power ($m=n=4$), so nothing can be excluded. Under our abstention rule both cases return **uncertified**, for different reasons, and neither is a tuning failure: the TOFU case is exactly what the identifiability analysis predicts for inferable facts (Section 9). The restoration-vs-suppression *diagnostic* against TOFU’s real retrained oracle is unaffected and remains the load-bearing TOFU claim.

9 Identifiability Limits of Oracle-Free Selection

We formalize when oracle-free selection is possible. Intuitively, a selector that never sees the retrained oracle can only succeed if the oracle’s notion of “adequately forgotten” is recoverable from oracle-free observables.

Setup A world $w \in \mathcal{W}$ specifies retained and forgotten data, an injected model $M_{\text{inj}}(w)$, and the retrained oracle $M_{\text{or}}(w)$. Let $\{M_a(w) : a \in \mathcal{I}\}$ be the candidate unlearned models, let d be a prescribed model-discrepancy measure, and define the candidate forget-NLL $q_a(w)$, the oracle forget level $\tau^*(w) := q_{\text{or}}(w)$, and

$$\mathcal{A}^*(w) := \{a \in \mathcal{I} : d(M_a(w), M_{\text{or}}(w)) \leq \varepsilon \text{ and } q_a(w) \geq \tau^*(w)\}.$$

Let ν_w denote the population law of NLLs assigned by $M_{\text{inj}}(w)$ to never-learned probes, and let $X(w) := (\{q_a(w)\}_{a \in \mathcal{I}}, \nu_w, \{r_a(w)\}_{a \in \mathcal{I}})$, where $r_a(w)$ is the round-trip residual of candidate a . An oracle-free selector is a deterministic map S satisfying $S(w) = s(X(w))$ for some s ; it is oracle-consistent on $\mathcal{W}$ if $S(w) \in \mathcal{A}^*(w)$ for every $w \in \mathcal{W}$. The threshold τ^* is identifiable from X on $\mathcal{W}$ if there exists a functional g with $\tau^*(w) = g(X(w))$ for all $w \in \mathcal{W}$.

Assumption 1 (Threshold separation and tie-breaking (TR)). Fix a deterministic ordering of $\mathcal{I}$ for resolving all ties. For any $w, w' \in \mathcal{W}$ with $X(w) = X(w')$ and $\tau^*(w) \neq \tau^*(w')$, threshold separation and the fixed tie-breaking convention yield $\mathcal{A}^*(w) \cap \mathcal{A}^*(w') = \emptyset$.

Assumption 2 (Residual soundness (RS)). For $t \in \mathbb{R}$, let $E_t(w) := \{a : q_a(w) \geq t\}$. For every w , $E_{\tau^*(w)}(w) \neq \emptyset$, and the tie-broken residual minimizer $\hat{a}(w) := \arg \min_{a \in E_{\tau^*(w)}(w)} r_a(w)$ satisfies $d(M_{\hat{a}(w)}(w), M_{\text{or}}(w)) \leq \varepsilon$.

Assumption 3 (Counterfactual exchangeability (CE)). There exists a known distributional functional Γ such that, for every $w \in \mathcal{W}$,

$$\tau^*(w) = \Gamma(\mathcal{L}(\text{NLL}_{M_{\text{or}}(w)} \text{ on forgotten examples})) = \Gamma(\mathcal{L}(\text{NLL}_{M_{\text{inj}}(w)} \text{ on never-learned probes})) = \Gamma(\nu_w).$$

Theorem 1 (Identifiability of oracle-free unlearning selection). *Under (TR) and (RS), an oracle-free oracle-consistent selector on $\mathcal{W}$ exists if and only if τ^* is identifiable from X on $\mathcal{W}$.*

Proof. For sufficiency, suppose $\tau^*(w) = g(X(w))$. Define $S(w) := \arg \min_{a: q_a(w) \geq g(X(w))} r_a(w)$ with the fixed tie-breaking order. Every quantity is determined by $X(w)$, so S is oracle-free. Since $g(X(w)) = \tau^*(w)$, (RS) gives $d(M_{S(w)}(w), M_{\text{or}}(w)) \leq \varepsilon$ and construction gives $q_{S(w)}(w) \geq \tau^*(w)$; hence $S(w) \in \mathcal{A}^*(w)$. For necessity, if τ^* is not identifiable there exist w, w' with $X(w) = X(w')$ but $\tau^*(w) \neq \tau^*(w')$. Any oracle-free selector returns the same index in both; by (TR) the adequate sets are disjoint, so that index cannot be adequate in both. Thus no oracle-free selector is oracle-consistent on $\mathcal{W}$. $\square$

Lemma 1 (Counterfactual sufficiency). *Under (CE), $\tau^*(w) = \Gamma(\nu_w)$ is identifiable from $X(w)$; consequently, under (RS), the selector $S(w) = \arg \min_{a: q_a(w) \geq \Gamma(\nu_w)} r_a(w)$ is oracle-consistent. Taking Γ to be the population mean yields a cutoff set by the never-learned distribution, with no hand-tuned margin.*

Proposition 2 (Failure for inferable facts). *Suppose $\mathcal{W}$ contains $w_{\text{cf}}, w_{\text{inf}}$ with $X(w_{\text{cf}}) = X(w_{\text{inf}})$ but $\tau^*(w_{\text{inf}}) < \tau^*(w_{\text{cf}})$, because retained data in w_{inf} reconstruct the forgotten fact whereas it is genuinely counterfactual in w_{cf} . Then τ^* is not identifiable from X ; under (TR), no oracle-free selector is oracle-consistent on any world-class containing both.*

Consequence. The certificate’s success on nonce facts and its boundary on TOFU are the *same* phenomenon (Proposition): TOFU is the predicted failure region, not an anomaly. There, matched never-learned probes sit far above the oracle’s own (inferability-driven) forget level, so (CE) fails. Constructing a retain-only reference to “fix” TOFU merely re-derives the oracle. This bounds oracle-free selection that leans on a never-learned reference, the mechanism studied here; it is not a universal impossibility for selectors with other access, such as white-box, adaptive-query, or known-pipeline methods.

Table 3: The identifiability result, operationally. The selector observes only X (never-learned-probe NLLs + candidate quantities), never the oracle; it succeeds exactly when the oracle’s forget level τ^* is recoverable from X .

element	role in the selector
observable X	never-learned-probe NLL law ν_w + candidate residuals/forget-NLLs (no oracle)
target τ^*	the oracle’s forget level; identifiable from X iff facts are counterfactual
forget-qualification	exclusion test: reject candidates below the never-learned reference
counterfactual facts	(CE) holds $\Rightarrow \tau^* = \Gamma(\nu_w) \Rightarrow$ oracle-free selection is possible
inferable facts	retained data reconstruct the target, (CE) fails $\Rightarrow$ no never-learned reference recovers τ^* (TOFU regime)

Measuring the headroom oracle-free. The identifiability condition is the counterfactual headroom $G = \overline{\text{NLL}}_{M_{\text{or}}}(F) - \overline{\text{NLL}}_{M_{\text{inj}}}(F)$: large when injecting F created behavior the retain lacks (identifiable), ≈ 0 when retain reconstructs F (inferable). We estimate it *without the oracle* by $\hat{G} = \overline{\text{NLL}}_{M_0}(F \mid \text{evidence}) - \overline{\text{NLL}}_{M_{\text{inj}}}(F)$, evaluating each fact with its supporting evidence prepended in context. On a controlled graded-inferability axis (each fact’s object made entailable from retain via alias chains over a nested support fraction; 5 levels $\times$ 5 seeds, a matched retraining reference per cell), $\hat{G}$ tracks the true headroom at **Spearman 0.97** (mean absolute error 1.7 nats). The theorem’s regime boundary is therefore measurable in deployment: $\hat{G}$ is high on counterfactual nonce facts and collapses toward zero on inferable facts. *Scope:* $\hat{G}$ detects the regime; in our controlled study it did *not* further predict within-regime certificate selection-regret (no headroom–regret contraction), so we use it only as a regime detector.

Assumptions (honest scope). (CE) is a strong counterfactual invariance claim and fails when retained data imply forgotten facts; (RS) and (TR) are strong ranking/separation assumptions, not consequences of oracle-free observability; and population identifiability does not imply finite-sample recovery; finite-probe implementations should use a data-derived confidence set rather than a fixed margin.

10 Related Work

Exact and certified unlearning. The retrained (“retain-only”) model is the accepted gold standard for forgetting. Exact-unlearning systems partition or checkpoint training so that removal is provable (Bourtoule et al., 2021), and a parallel line gives *certified* removal under convexity or bounded-influence assumptions (Guo et al., 2020; Neel et al., 2021; Sekhari et al., 2021). These guarantees do not transfer to full-parameter language models; rather than certify deletion, we construct a dose-matched retraining reference in a controlled testbed and ask whether good unlearning can be *selected* without it.

Unlearning methods for language models. Approximate methods optimize forget/retain objectives directly: gradient ascent/difference and negative preference optimization (Zhang et al., 2024), scrubbing (Kurbanji et al., 2023), selective synaptic dampening (Foster et al., 2024), representation misdirection (RMU, introduced with WMDP) (Li et al., 2024), targeted concept removal (Eldan & Russinovich, 2023; Liu et al., 2024), and unlearning of pretrained LLMs (Yao et al., 2024). We show that such forget/retain scores can be driven by *suppression* that moves the model *away* from the retrained reference, whereas restoration-style methods (task-vector negation, path-specific reversion) stay closer to it.

Benchmarks and evaluation. TOFU (Maini et al., 2024), MUSE (Shi et al., 2025), WMDP (Li et al., 2024), and unified suites (Dorna et al., 2025) score forget/retain behavior against fixed references. We add a dose-matched retraining reference that makes “closeness to the retrained model” a directly measurable quantity, and use it to study oracle-free *selection* rather than to propose another metric.

Recovery, relearning, and leakage. Apparent forgetting is frequently reversed by quantization (Zhang et al., 2025) and exposed by black-box probing (Doshi & Stickland, 2024); membership-inference (Shokri et al., 2017; Carlini et al., 2022) and extraction (Carlini et al., 2021) attacks quantify residual leakage. Our forget-qualified round-trip residual (delete, then re-acquire) is a selection signal in this spirit: it probes whether the forgotten distribution is genuinely *restorable* rather than merely hidden.

Task/model arithmetic and merging. Task arithmetic (Ilharco et al., 2023), model soups (Wortsman et al., 2022), and Fisher-weighted merging (Matena & Raffel, 2022) manipulate models in weight space. We repurpose task-vector negation as a strong *restoration* exemplar rather than an editing tool.

Knowledge editing. ROME (Meng et al., 2022a), MEMIT (Meng et al., 2022b), MEND (Mitchell et al., 2022a), and SERAC (Mitchell et al., 2022b) locate and edit factual associations. We borrow the controlled-fact methodology (injecting nonce, counterfactual facts) but target *deletion-quality measurement* against a matched retraining reference rather than editing.

Causal identifiability. Our identifiability result is stated in the language of causal identifiability and abstraction (Peters et al., 2017; Beckers & Halpern, 2019): the counterfactual-exchangeability condition that governs when oracle-free selection is oracle-consistent is an identifiability condition on the forget-qualification threshold, not a claim of full causal-model recovery.

Delta. We contribute a matched-retraining *audit methodology* (run the retraining reference itself through every proposed criterion, with replication noise quantified) and its findings: the trained-probe criterion rewards residual knowledge; absolute certification thresholds fail their own reference; a base-anchored held-out screen survives as a validated *necessary* test with measured true-accept, sensitivity, and a drift-seam detection property; a damage-relative recalibration yields a selective partial positive with selections inside retraining noise; and the adversarial boundary of any forward-only battery is measured, not assumed. Two

limits jointly scope the enterprise: an identifiability limit (which facts admit an oracle-free forget threshold at all) and a finite-sample certifiability limit (when the screen can reject at all). The resulting guarantee is a *selective* test for methods-as-produced under stated assumptions, not a formal deletion certificate, and it abstains when any limit binds.

11 Limitations

- **Threat model.** Certification is an empirical selective test for *methods-as-produced*: Section 6 measures that behavioral adversaries (a fixed-magnitude logit suppression attack) defeat the full forward battery in 12/45 cells, and an adaptive bilevel attack on the round-trip residual is untested. We quantify this boundary rather than claim past it.
- **Retrospective, veto-less recalibration.** The damage-relative certificate is evaluated retrospectively on stored candidates, without the relearn-veto axis (no stored relearning trajectories); a prospective run with all axes is future work.
- **Small forget set.** $m = 4$ forget facts per cell (with $n = 16$ probes); the screen’s resolution clears Holm’s threshold, but power against small effect sizes is limited, and a 16-fact breadth panel was designed and deliberately deprioritized in favor of the audit runs.
- **Tolerance concentration and few redraws.** Per-family δ_f spans 0.82–2.92 nats, is estimated from only three retraining redraws (so it is itself uncertain), and most certifications arise in the noisiest-replication family; the global-minimum sensitivity row is reported alongside, and a coverage-calibrated tolerance from more redraws would sharpen the certified counts.
- **Scope of the impossibility.** Our finite-query argument concerns *oracle-free, forward-only, black-box* protocols over an unrestricted candidate class; it does not preclude certification under white-box access, known training pipelines, restricted method classes, or adaptive/randomized querying, which are open. The logit-suppression evasion is the empirical face of this scope, not a universal impossibility.
- Selection is oracle-free only for **counterfactual** facts (Theorem 1); inferable-fact benchmarks (TOFU) need the reference, though the diagnostic still applies there.
- Small models ($\leq 1.7\text{B}$ in the matrix; Phi-1.5 for TOFU); nonce facts are synthetic by design (to make the matched reference constructible); TOFU provides the realistic corroboration. bf16 training nondeterminism means every reference-derived number carries the replication noise quantified in Section 5.

12 Conclusion

Good unlearning is distribution *restoration* toward the retrained model, not answer suppression. Evaluating it demands instruments that survive their own audit. Ours did not all survive: the trained-probe criterion rewards residual held-out knowledge, and the absolute certificate bar fails the retraining reference itself, in most cells, for reasons (threshold mis-scaling; a budget-driven round-trip floor) that no candidate could overcome. What survives the audit is precise. A base-anchored held-out screen is a validated *necessary* test: perfect sensitivity on the injected model, high true-accept on the reference, and a measured partial detection of routing through a distributional seam. A damage-relative recalibration, frozen before analysis, turns the audit into a selective partial positive: certification in a third of cells, abstention elsewhere, and selections within retraining noise of the reference where the common trained-probe criterion misses by five nats. And the boundary is quantified rather than assumed: a fixed-magnitude suppression attack defeats this forward-only battery in a quarter of cells, which is the empirical face of the identifiability and finite-query limits that scope oracle-free selection. A certificate worth the name must be able to answer “uncertified” — and must itself be certifiable by the reference it appeals to. We report ours both ways.

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A Reproducibility

Every model used is open-weight and named with its exact checkpoint in Section 2, and the artifacts needed to audit the claims in this paper are specified here rather than deferred to an external repository. The injection and re-acquisition budgets, the method-side compute budget, the candidate grids, the never-learned probe construction, and the evaluation protocol are given in Appendix B; together with the reference construction of Section 2 these are sufficient to reimplement the testbed, the matched retraining reference, and the forget-qualified round-trip selector from the paper alone. The recalibration rules of Section 7 and the challenge-panel constructions of Section 6 were frozen and committed before their analyses ran, the sealed AUDIT template pool was scored exactly once, and every model-seed cell is reported; every headline number in the paper is produced by a committed analysis script. To let others verify that the sealed pool was fixed in advance

(the property that makes the challenge-panel numbers independent), we publish a cryptographic hash of the AUDIT pool with this paper and release the pool itself only after decisions, so the screen validation is reproducible without compromising the one-time-use guarantee. Code and configuration to regenerate every table and figure will be released publicly; we will add the repository link and the hash to this section in a revision.

B Methods and Hyperparameters

Injection and oracle. Facts are nonce (subject, relation, object) triples generated from disjoint random seeds; each is injected by continued pretraining with an interleaved wikitext retain mix (ratio 3:1 retain:fact per step). Controlled runs use $|F| = |R| = 4$ facts, diversity $K = 3$ phrasings, 200–250 injection steps, learning rate 2×10^{-5} , gradient clipping 1.0. M_{inj} co-acquires F then R ; the oracle M_{or} replays the identical stream with F removed; a forget-only model M_{Fonly} supports task-vector negation.

Unlearning candidates. From M_{inj} : (i) *KL reversion* (full-parameter): match M_0 on F probes and M_{inj} on retain, retain weight $w \in \{3, 12, 50, 200\}$; (ii) *gate_proj mask* (SwiGLU only): learnable binary keep/zero mask under the same KL objective, sparsity 0.1, α ramp $10^2 \rightarrow 10^4$; (iii) *task-vector* $M_{\text{inj}} - c(M_{\text{Fonly}} - M_0)$, $c \in \{0.5, 1, 1.5, 2\}$; (iv) *GA+retain* and (v) *NPO+retain* ($\beta = 0.1$), learning rate 5×10^{-6} , $w \in \{3, 12, 50, 200\}$. Method budget ≈ 110 –120 steps; matched compute across methods.

Certificate. Round-trip residual = $\text{KL}(\text{reacquire}_F(C) \parallel M_{\text{inj}})$ on retain probes + wikitext, with 75–80 reacquisition steps (5 logged chunks). Never-learned probes are $n = 16$ facts of the same nonce type from unused seeds that no model ever saw in training. The under-forgetting exclusion test is a one-sided Mann–Whitney U of the candidate’s $m = 4$ base-anchored per-fact forget deltas against its 16 probe deltas (held-out templates averaged within fact before ranking), Holm-corrected at preregistered $\alpha = 0.05$.

Audit and panel. Template pools are disjoint by construction and unit-tested: injection (6 families), unlearning loss (cloze + injected), evaluation (4 held-out families), calibration (4, never used here), and a sealed AUDIT pool (6, scored once, for the challenge panel). Reference replication: 3 independent re-training draws in one representative cell per family; $\delta_f = \text{P90 per-fact held-out delta gap between draws}$; $\varepsilon_f = \text{draw-to-draw spread of retain-NLL and round-trip residual}$. Round-trip budget curve: self-closure of M_{inj} at $\{15, 30, 45, 75\}$ reacquisition steps. Challenge panel: logit suppression = -10 additive penalty on forget-answer token logits; entity router dispatches on exact forget-subject token-sequence match; embedding corruption zeroes all subject/object token rows; interpolations at $\lambda \in \{0.25, 0.5, 0.75\}$. Recalibration rules (thresholds, margins, degeneracy gate, selection rule) were frozen and committed before the analysis ran; all analysis scripts are committed and every reported number is regenerable from stored per-cell artifacts.

Metrics. oracle-KL = mean $\text{KL}(C \parallel M_{\text{or}})$ over $F+R$ answer spans; forgetting = held-out paraphrase NLL on F ; retain = cloze NLL on R . The frozen matrix is 9 configurations $\times 5$ seeds over five families (Qwen2.5-0.5B/1.5B, SmoLLM2-360M/1.7B, Pythia-410M/1.4B, GPT-2-medium/large, OLMo-2-1B); the rule is fixed before evaluating held-out families.

TOFU. forget10/retain90 on Phi-1.5; $M_0 = \text{phi-1.5}$, $M_{\text{inj}} = \text{tofu_ft_phi-1.5}$, $M_{\text{or}} =$ the official `tofu_ft_retain90_phi-1.5` retrained reference; 40 forget and 100 retain items. All experiments run on a single 48 GB RTX 6000 (Ada).